IVF Digital Twin

Version 6.2

An Integrated Multi-Source Ensemble Platform for Stage-Stratified IVF Outcome Prediction

Stochastic Simulation · Neural Network Ensembling · Bayesian Conditional Updating Unsupervised Phenotype Classification · Diffusion-Based Generative Module

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For use with the open-source repository · Please read before running the model

Abstract

Current clinical decision-support tools for in vitro fertilization (IVF) typically address isolated endpoints — oocyte yield, embryo morphology, or implantation probability — and return point estimates without quantifying the substantial biological variability inherent to reproductive medicine. They cannot incorporate stage-by-stage information as the cycle unfolds, do not benchmark a given patient against the spectrum of documented IVF protocol phenotypes, and do not integrate independent generative verification of laboratory outcome distributions.

We constructed a five-layer probabilistic prediction system combining: (i) a seven-stage stochastic Monte Carlo pipeline propagating full probability distributions through controlled ovarian stimulation, oocyte retrieval, maturation, fertilization, blastulation, good-quality blastocyst formation, age-stratified euploidy, and post-thaw survival; (ii) a per-transfer pregnancy ensemble combining logistic regression with a KPIScore-based laboratory performance metric; (iii) a Kolmogorov-Arnold Network and FT-Transformer neural network ensemble (KAT) with Venn-Abers conformal calibration; (iv) a Beta-Binomial Bayesian posterior synthesizing clinical prior, retrospective clinic data, and neural network output; (v) an unsupervised nearest-centroid cluster classifier based on three published IVF protocol phenotypes; and (vi) a diffusion-based generative module (CSDI Hybrid v3) that independently reconstructs the embryological outcome distribution from approximately 15,000 clinical cycles without parametric rate assumptions. All layers were validated on retrospective cohorts from independent centers.

On a retrospective validation cohort, the stochastic pipeline reproduced observed median oocyte and blastocyst counts with Spearman correlations of 0.73 and 0.41 respectively. Pregnancy prediction achieved an AUC of 0.63 with a Brier score of 0.22. The CSDI Hybrid generative module achieved an AUC of 0.66, Brier score of 0.21, and Expected Calibration Error of 0.029 (predicted vs. actual prevalence deviation: 0.1 percentage points) on a hold-out test set of 1,520 cycles. The unsupervised cluster classifier correctly assigned all five reference scenarios to their expected phenotype.

The IVF Digital Twin is the first IVF prediction system to integrate stochastic stage-wise simulation, multi-source per-transfer ensemble, calibrated neural-network prediction, Bayesian evidence synthesis, mid-cycle conditional updating, unsupervised phenotype classification, and data-driven diffusion generative modelling within a single transparent, auditable framework. All coefficients are traceable to peer-reviewed sources, and the architecture supports prospective external validation.

Document Structure

1. Introduction

In vitro fertilization (IVF) is among the most technologically sophisticated and clinically uncertain domains of modern medicine. More than 2.5 million treatment cycles are performed worldwide each year, yet live birth rates per cycle remain constrained to 25–35%, and the patient experience is characterized by substantial psychological, physical, and financial burden. The fundamental limitation is not an absence of predictive tools — many exist — but the fragmented, point-estimate-focused, and biologically incomplete character of available systems.

Existing IVF prediction systems address isolated endpoints. The CDC IVF Success Estimator and the OPIS calculators report cumulative pregnancy probabilities without decomposing the prediction into underlying biological transitions. The Orchid Embryo Banking Calculator provides stage-specific point estimates without confidence intervals or downstream propagation of uncertainty. The Herasight framework introduces distribution-based stage-structured modelling with conditional updating but does not quantify OHSS risk or benchmark patients against documented IVF protocol phenotypes.

Most commercial calculators produce single scalar probabilities, collapsing the cascade of conditional events — oocyte retrieval, maturation, fertilization, blastulation, euploidy, transfer, implantation — into one number. They obscure the very questions that patients and clinicians most need answered: what is the full range of plausible outcomes? What is the risk of cycle cancellation? What happens if retrieval underperforms? How will mid-cycle observations change the prognosis?

Recent consensus in medical artificial intelligence emphasizes that high-stakes decisions require uncertainty-aware, transparent models that propagate variability throughout the prediction pipeline. The Monte Carlo method was selected as the computational foundation of this platform because an IVF cycle is not a deterministic sequence but a chain of stochastic biological filters, and the quantities of clinical interest are therefore not point estimates but full probability distributions. Analytical propagation of uncertainty through such a cascade — a convolution of a zero-inflated negative-binomial oocyte count with a sequence of binomial filters, age-dependent blastulation and euploidy rates, and a logistic implantation step — has no closed-form solution. Monte Carlo simulation resolves this by sampling the entire cascade many thousands of times, allowing the joint distribution of every downstream quantity to emerge directly.

We therefore designed and developed the IVF Digital Twin: an integrated, AI-augmented stochastic prediction platform that models the entire IVF treatment trajectory as a sequential probabilistic pipeline. This manuscript describes its final integrated form (version 6.2), which adds a fifth architectural layer — a diffusion-based generative module trained on approximately 15,000 clinical cycles — to provide independent data-driven corroboration of the parametric Monte Carlo predictions.

2. System Overview

The IVF Digital Twin is organized into five conceptual layers that operate in series during a single prediction run (Table 1). Each layer addresses a distinct epistemic question and contributes a distinct piece of information to the final prediction.

Table 1. Five-layer architecture of the IVF Digital Twin v6.2

Layer	Function	Key methods
L1 — Stochastic pipeline	Propagates full probability distributions through seven biological stages of the IVF cycle, from stimulation through post-thaw survival.	Truncated Normal; logistic-binomial filters; Beta-binomial; age-stratified euploidy table. Coefficients from ART-ONE, Herasight, Romanski, Sainte-Rose, Franasiak, Armstrong, Coello.
L2 — Per-transfer ensemble	Combines two independent per-transfer pregnancy estimates: a logistic regression on clinical predictors and a KPIScore-based laboratory performance metric.	FORTUNE-structure logistic; KPIScore (5–25) with per-integer Beta-distributed 95% CIs; logit-weighted ensemble; three-level pregnancy decomposition.
L3 — Neural network ensemble + Bayes	Calibrated final-layer prediction by a KAN + FT-Transformer ensemble. Bayesian Beta-Binomial posterior fuses NN output, clinical prior, and retrospective data.	KAN (3-layer, B-spline activations) + FT-Transformer; Venn-Abers conformal calibration; NVSA correction; covariate-dependent Beta regression prior; per-attempt decay curve.
L4 — Cluster classifier	Independent unsupervised assessment via nearest-centroid assignment to three published IVF protocol phenotypes (Poor, Standard, High responder).	Nearest-centroid in z-score-standardised 18-dimensional space; centroids from 1,556 cycles, k-means k=3; PCA visualization with synthetic clouds.
L5 — Diffusion generative module	Independently reconstructs embryological outcome distributions directly from clinical data. Provides calibrated pregnancy probability and conformal prediction intervals.	CSDI Hybrid v3: Transformer-based diffusion model trained on ~15,000 cycles; LightGBM + Platt classifier; split conformal prediction with coverage guarantees.

All five layers share a single Monte Carlo loop of N = 5,000 iterations per patient. Within each iteration, the patient's demographic and ovarian reserve inputs are propagated through the stochastic pipeline (L1), simultaneously feeding the per-transfer ensemble (L2), the neural network (L3), and the cluster classifier (L4). The diffusion module (L5) then receives the pipeline medians and independently generates its own distribution of embryological trajectories.

The architecture deliberately combines complementary epistemic sources. The stochastic pipeline encodes population-level biological knowledge from large registries. The FORTUNE-like component reflects an established clinical prediction model derived from over 3,000 patients in randomized trials. The neural network captures complex, possibly non-linear, interactions that pre-specified linear models cannot. The Bayesian posterior re-anchors the prediction to the clinic's own historical pregnancy frequency. The cluster classifier provides unsupervised phenotype labelling. The diffusion module provides an entirely data-driven, assumption-free verification from a different modeling paradigm. Each layer contributes information that the others do not, and the explicit ensembling makes the contribution of each layer auditable.

3. Stochastic Pipeline (Layer 1)

The stochastic pipeline models an IVF cycle as a sequence of seven probabilistic stages. Each stage receives as input the output of the preceding stage and produces a full probability distribution rather than a point estimate. Patient inputs are: female age (years), anti-Müllerian hormone (AMH, ng/mL), antral follicle count (AFC), and body-mass index (BMI, kg/m^2).

3.1 Stage 1 — Retrieved oocytes

The number of retrieved oocytes (OKK) is modelled using a Zero-Inflated Negative Binomial (ZINB) distribution. This choice addresses two fundamental limitations of the truncated Normal used in earlier platform versions. First, a proportion of stimulated cycles end before oocyte retrieval — due to poor ovarian response, premature ovulation, or protocol cancellation — producing a structurally distinct mass of zero-yield cycles that cannot be captured by any unimodal continuous distribution. Second, oocyte yields are right-skewed with heavier tails than the Normal, particularly among high responders: variance grows quadratically with the mean rather than linearly. Both properties are formally accommodated by the ZINB.

The model has two components. The zero-inflation component assigns each iteration a probability of cancellation via a logistic function of patient covariates:

$$\mathrm{logit}(P\_zero) = -3.2 + 0.40 \cdot z\_age - 0.60 \cdot z\_AMH - 0.80 \cdot z\_AFC$$

where $z\_age = (age - 36.3)/5.5$, $z\_AMH = (AMH - 2.0)/1.2$, $z\_AFC = (AFC - 12)/9$. Calibration against published cancellation rates yields clinically plausible zero-probabilities: approximately 2–3 % for a standard 35-year-old with AMH 2.5 ng/mL and AFC 15; 18 % for a 42-year-old with AMH 0.5 ng/mL and AFC 6; and below 0.3 % for a 28-year-old high responder with AMH 4.5 ng/mL and AFC 28. These figures are consistent with the zero-component coefficients reported by Craig et al. for the HFEA registry (positive age coefficient +0.050, stimulation coefficient −0.796 on the zero logit).

The count component, applied only to non-cancelled iterations, follows a Negative Binomial distribution with mean μ derived from the ART-ONE linear predictor and dispersion parameter $\theta = 5.0$ calibrated to the HFEA negative binomial fit of Craig et al:

$$\mu = 3.25 + 1.20 \cdot AMH + 0.55 \cdot AFC - 0.15 \cdot age - 0.03 \cdot BMI$$

The NB dispersion $\theta = 5.0$ gives variance $= \mu + \mu^2/\theta$, which grows quadratically with the mean. For $\mu = 9$ (standard responder) this yields SD ≈ 5.0, compared with SD = 3.2 from the former linear formula — correctly reflecting the greater spread observed in registry data. For $\mu = 19$ (high responder) the NB SD reaches 9.6, with the 5th–95th percentile interval [6, 37], capturing the full range of high-responder outcomes that a Normal distribution with SD = 5.4 systematically underestimates. Oocyte counts are clipped to [0, 50].

The zero-inflation component assigns each iteration a cancellation probability via a logistic function of patient covariates, yielding approximately 2–3% for a standard 35-year-old with AMH 2.5 ng/mL and AFC 15; 18% for a 42-year-old with AMH 0.5 ng/mL and AFC 6; and below 0.3% for a 28-year-old with AMH 4.5 ng/mL and AFC 28. The count component follows a Negative Binomial with mean derived from the ART-ONE linear predictor and dispersion calibrated to the HFEA registry. Representative outputs: standard responder (age 35, AMH 2.5, AFC 15): mean 8.4, SD 5.0; poor responder (age 42, AMH 0.5, AFC 6): mean 0.4, cancelled-cycle rate 18.2%; high responder (age 28, AMH 4.5, AFC 28): mean 19.0, SD 9.6.

3.2 Stage 2 — Mature (MII) oocytes

Maturation is modelled as a binomial filter applied to the simulated oocyte count, with a per-oocyte success probability obtained from a logistic regression. The coefficients are taken verbatim from the Herasight simplified maturity model (Table A1, column 3), fit on 90,479 HFEA cycles. For our clinical context (stimulation = 1, ICSI, unexplained infertility) the linear predictor reduces to:

$$\text{logit}(p\_MII) = 2.4665 + 0.005\cdot\text{age} - 0.782\cdot\text{stim} + 0.24\cdot\text{AMH} - 0.069$$

This collapses to a population-mean maturity rate of approximately 90.7 percent at the HFEA average age and AMH, which is precisely the empirical mean reported by the source registry. The MII count for iteration i is then drawn from Binomial (OKK_i, p_MII). The choice of fixed-coefficient logistic over earlier beta-binomial heuristics eliminates spurious overdispersion and ensures that the maturity stage is traceable to a single peer-reviewed source.

3.3 Stage 3 — Two-pronuclear zygotes (2PN)

Fertilization is modelled analogously. The per-MII success probability is taken from the Herasight simplified fertilization model (Table A1, column 5), with the same domain reduction (ICSI, stimulation, unexplained infertility):

$$\text{logit}(p\_fert) = 1.1678 + 0.004\cdot\text{age} - 0.303\cdot\text{stim} - 0.051$$

This yields a population-mean fertilization rate of 72.3 percent at the HFEA mean age, matching the empirical value. The conventional-IVF coefficient (−0.296) is omitted because all cycles in our clinic use ICSI. The 2PN count is drawn from Binomial (MII_i, p_fert). Stages 2 and 3 together replace earlier expert-driven heuristic models with peer-reviewed registry-derived coefficients while preserving the upstream stochasticity of the oocyte yield distribution.

3.4 Stage 4 — Blastocysts

Blastocyst formation is modelled as a binomial filter with an age-dependent rate. Earlier versions of the platform used a simple linear age effect (0.62 − 0.006·age), which substantially under-estimated blastulation among patients younger than 35. The current v6.1 formulation, calibrated against the data of Romanski et al. (3,362 patients) and Sainte-Rose et al. (4,952 zygotes), holds the blastocyst formation rate stable through age 40 and applies an accelerated decline thereafter:

$$\text{p\_blast} = \text{clip}(0.70 - 0.012 \cdot \max(0, \text{age} - 40),\ 0.30,\ 0.75)$$

To reflect inter-cycle and inter-laboratory variability that fixed coefficients cannot capture, the rate for each Monte Carlo iteration is itself drawn from a Normal distribution with mean p_blast and standard deviation 0.06, clipped to [0.10, 0.90]. The blastocyst count is then drawn from Binomial (2PN_i, p_blast,i). The revised coefficients produce age-specific blastulation rates of 70 percent for ages 21–40 and 64 percent at age 45, in close agreement with the 60–67 percent stable plateau and post-40 decline documented in the source literature.

3.5 Stage 5 — Good-quality blastocysts

The proportion of blastocysts graded as good quality (Gardner 3BB or higher) is modelled as another binomial filter with an age-dependent rate. The updated v6.1 coefficients reflect the observation that good-quality fraction is high among younger patients and declines gradually with maternal age:

$$\text{p\_good} = \text{clip}(0.78 - 0.008 \cdot \max(0, \text{age} - 35),\ 0.40,\ 0.85)$$

As in Stage 4, the per-iteration rate is sampled from a Beta distribution with concentration parameter 10 around the central mean, capturing grading variability across embryologists and laboratories. The good-quality count is drawn from Binomial (blasts_i, p_good,i). At the population mean of 35 years the central rate is 0.78, declining to 0.72 at age 42 and to 0.65 at age 48. These values match the empirical good-blast fractions reported in independent clinical analyses and align with the Herasight per-patient calculator output (5–6 high-quality blastocysts at age 21 from 12 retrieved oocytes).

3.6 Stage 6 — Euploid embryos

Aneuploidy is the single largest determinant of implantation failure and miscarriage, and its prevalence rises sharply with maternal age. The age-specific euploidy rate is taken from the combined evidence of Franasiak et al. (15,169 trophectoderm biopsies) and Armstrong et al. (86,208 cycles) and from our routine practice with 4863 treatment cycles, synthesized into an age-stratified central value.

To convert this single central value into a per-iteration random rate, the central value is used as the mean of a Beta (α, β) distribution with concentration $\kappa = 6$ ($\alpha = \text{mean}\cdot\kappa$, $\beta = (1 - \text{mean})\cdot\kappa$). Each individual blastocyst is then declared euploid or aneuploid via Bernoulli trials with the sampled probability. This formulation introduces appropriate variability at the embryo level while preserving the age-stratified central tendency. For each Monte Carlo iteration the simulated euploid count is bounded above by the good-blastocyst count.

Age range	Central euploidy rate
< 30 years	0.70
30–34 years	0.65
35–37 years	0.55
38–39 years	0.35
40–41 years	0.18
≥ 42 years	0.10

3.7 Stage 6b — Post-thaw survival

Following preimplantation genetic testing for aneuploidy (PGT-A), euploid blastocysts are vitrified and warmed before transfer. Survival after warming is modelled as a Binomial filter with a fixed 95 percent per-blastocyst survival rate, in line with the multicentre data of Coello et al. (2021). This stage is short and deterministic in its rate but stochastic in its count, preserving the propagation of distributional uncertainty.

3.8 Summary of the stochastic pipeline

The pipeline is fully chained: the output of each stage is the trial count for the next. Because every stage maintains a full N-iteration distribution, downstream metrics — euploid embryo count, OHSS risk, empty-cycle risk, pregnancy probability — inherit the upstream variability faithfully. The pipeline can be conditioned on observed values at any stage (Section 8), replacing the relevant random variable with a point mass and propagating it deterministically through earlier stages and stochastically through later ones.

4. Monte Carlo Engine and Risk Quantification

4.1 Sample size and convergence

The Monte Carlo engine draws N = 5,000 independent samples per patient at each pipeline stage. This sample size was chosen empirically: at N = 1,000 the standard error of cumulative pregnancy probability estimates exceeded 1.0 percentage points across replicate runs, whereas at N = 5,000 it stabilises below 0.4 percentage points, well within the resolution required for clinical counselling. Above 10,000 samples the marginal gain in precision is negligible while computation time grows linearly. All Monte Carlo runs use a deterministic seed when reproducibility is required and an entropy-based seed otherwise; the random-number generator is the NumPy Mersenne Twister implementation.

4.2 Explicit risk metrics

4.2.1 Ovarian hyperstimulation syndrome (OHSS)

Following the ESHRE 2023 OHSS guideline and the meta-analysis of Humaidan et al., three operational risk categories are derived from the simulated oocyte distribution: Mild risk (retrieved oocytes ≥ 15); Moderate risk (15 ≤ oocytes < 20); Severe risk (oocytes ≥ 20). The empirical fraction of Monte Carlo iterations meeting each criterion directly informs the choice of ovulation trigger, the decision to proceed with fresh versus freeze-all transfer, and the intensity of post-retrieval monitoring.

4.2.2 Empty cycle risk

Two empty-cycle metrics are reported: the probability of obtaining zero blastocysts, and the probability of obtaining zero good-quality blastocysts. Both are computed as the empirical fraction of Monte Carlo iterations meeting the criterion. Neither metric is computed by any other publicly available IVF prediction system, yet both describe the worst-case scenario in which the patient receives no transferable embryo despite undergoing the full stimulation and retrieval procedure.

5. Pregnancy Probability — Multi-Source Ensemble (Layer 2)

The platform predicts clinical pregnancy at three nested probability levels, computed from a logit-scale ensemble of two independent per-transfer probability estimators.

5.1 FORTUNE-based per-transfer probability

The first per-transfer estimator follows the structure of the FORTUNE prediction model recently published by IVIRMA-NJ, applied to euploid warmed blastocyst transfers in top-decile SART clinics. The model expresses the log-odds of clinical pregnancy as a linear function of standardized patient covariates:

$$\mathrm{logit}(p\_F) = 0.40 - 0.55 \cdot z\_age + 0.15 \cdot z\_AMH - 0.20 \cdot z\_BMI$$

where z_age = (age − 36.3)/5.5, z_AMH = (log(AMH + 0.1) − log(2.1))/1.2, and z_BMI = (BMI − 24.0)/4.2. The intercept of 0.40 corresponds to a baseline per-transfer pregnancy probability of approximately 60 percent at the population mean, anchored to SART top-decile euploid-transfer live birth rates [25]. The age slope is intentionally weaker than in age-only models because much of the age effect is already encoded upstream through stage-by-stage euploidy decline. A small Gaussian random effect (SD = 0.07 on the logit scale, equivalent to ±3 percentage points around the central rate) is added per Monte Carlo iteration to reflect biological variability that fixed coefficients cannot capture.

5.2 KPIScore-based per-transfer probability

The second estimator is derived from KPIScore, a composite five-component laboratory index developed and validated in prior work. The score is calculated per Monte Carlo iteration using the simulated lab outputs of that specific iteration, so it is fully integrated with upstream stochasticity. Each of five components contributes 1, 3, or 5 points:

Component	Threshold for score 1	Threshold for score 3	Threshold for score 5
Female age (years)	≥ 40	37–39	≤ 36
AMH (ng/mL)	< 1	1.0–1.99	≥ 2
MII oocytes (simulated)	≤ 3	4–6	≥ 7
Fertilisation rate (%)	< 50	50–65	> 65
Good blastocysts (simulated)	0	1–2	≥ 3

Total scores range from 5 (worst) to 25 (best). Each integer score is mapped to a clinical implantation probability with published 95% confidence intervals, and per-iteration KPI probability is sampled from a Beta distribution moment-matched to the score's CI.

5.3 Logit-scale ensemble combination

The two per-transfer probability vectors (p_F and p_KPI) are combined per Monte Carlo iteration on the logit scale, following standard practice in stacked model averaging [26]:

$$\mathrm{logit}(p\_ensemble) = (1 - w)\cdot\mathrm{logit}(p\_F) + w\cdot\mathrm{logit}(p\_KPI)$$

The KPI weight w defaults to 0.5 (equal weighting). Setting w = 0 produces the pure FORTUNE prediction; w = 1 produces the pure KPI prediction. Logit-scale averaging is preferred over probability-scale averaging because it respects the natural odds-ratio metric of probabilities and avoids systematic bias near the 0/1 boundaries. The ensembled per-transfer probability is the central quantity used downstream by Layers 3 and 4.

5.4 Three-level pregnancy decomposition

Version 6.2 reports pregnancy at three explicitly ordered levels to resolve the apparent paradox that for poor responders, per-transfer rate can exceed cumulative cycle probability. Strict mathematical ordering is guaranteed at every level: per-transfer is always less than or equal to cumulative-if-viable (because additional transfers can only increase the probability of at least one success), and overall is always less than or equal to cumulative-if-viable (because P(viable) ≤ 1). A defensive max() safeguard at the algorithmic level prevents rare Monte Carlo noise from violating the ordering. The decomposition has three immediate clinical applications. First, it lets the clinician answer the per-transfer question directly when counselling a patient who is about to receive an embryo. Second, it lets the clinician quote a meaningful cycle-success number that does not depend on whether the patient happens to have viable embryos. Third, it makes the role of the empty-cycle risk transparent in the overall figure.

Level	Conditional event	Clinical meaning
[1] Per-transfer rate	Given that a transfer takes place	Probability of clinical pregnancy per single embryo transfer
[2] Cumulative-if-viable	Conditional on cycle producing ≥1 transferable embryo	Probability of ≥1 pregnancy over all planned transfers in this cycle
[3] Overall cycle success	From start of stimulation, including empty-cycle risk	Marginal probability of ≥1 pregnancy from the moment the patient begins treatment

5.5 Discrete pregnancy count distribution

Beyond the binary did-she-or-did-she-not-get-pregnant outcome, the platform reports the full discrete distribution over the number of clinical pregnancies achieved during sequential transfer of all available warmed euploid embryos. For each Monte Carlo iteration, the number of pregnancies is drawn from Binomial (n_transfers_i, p_ensemble,i). The probability of achieving at least k pregnancies is then computed empirically as the fraction of iterations meeting that threshold. This discrete distribution lets the clinician make precise statements such as P(at least one pregnancy this cycle) = 72 %, P(at least two) = 41 %, P(at least three) = 18 %, which are particularly relevant for patients pursuing family-building strategies that target multiple children from a single retrieval.

6. Neural Network Final Layer — The KAT Ensemble (Layer 3)

The third architectural layer is a calibrated neural-network ensemble called KAT (Kolmogorov-Arnold network + FT-Transformer). Positioned at the end of the prediction pipeline, it consumes all 18 simulated and observed laboratory variables from every Monte Carlo iteration and produces a per-iteration pregnancy probability that is subsequently NVSA-adjusted, Bayesian-synthesized, and projected onto the per-attempt decay curve.

6.1 Rationale

The pipeline up to and including Layer 2 is composed of linear, log-linear, or additive components. While transparent and well-validated, this structure cannot capture complex interactions among predictors — for example, the prognostic effect of three good-quality blastocysts may depend non-linearly on the patient's age, attempt number, and fertilization rate together. A neural-network layer provides a principled way to capture such interactions without pre-specification.

Two architectures with complementary strengths are ensembled. Kolmogorov-Arnold Networks (KANs) are based on the theorem that any multivariate continuous function can be represented as a composition of univariate functions, giving them an inductive bias toward smooth, interpretable decompositions that generalize well in small-data regimes. Feature-Tokenizer Transformers (FT-

Transformers) tokenize each feature into a learned embedding and apply self-attention across the resulting token sequence, giving them a strong bias toward capturing cross-feature interactions explicitly.

6.2 Architecture and calibration

The KAN component is implemented as a three-layer perceptron with B-spline activation functions (grid size 5, spline order 3), optimized with Adam and both L1 and entropy regularization to encourage parsimony. The FT-Transformer projects each of 18 input features into a learnable embedding and processes the token sequence with multi-head self-attention, followed by a classification head.

Group	Features
Patient demographics	Female age (years); IVF attempt number
Ovarian response	Follicle count on retrieval day; retrieved oocyte (COC) count
Embryological counts	MII oocytes (inseminated); 2PN zygotes; cleaving embryos on day 3; total blastocysts; good-quality blastocysts; day-5 embryos; cryopreserved embryos; transferred embryos
Laboratory KPIs	Fertilisation rate (2PN ÷ MII); cleavage rate (cleaving ÷ 2PN); blastocyst formation rate (blasts ÷ 2PN); TGBDR (good blasts ÷ 2PN); oocyte retrieval efficiency (COCs ÷ follicles)
Composite	KPIScore (5–25, computed inside the NN itself using follicle count as the second component)

The KAN and FT-Transformer are combined by a learned weighted average of their probability outputs, with two non-negative scalar weights (initialized at 0.5 each, learned jointly with the base models). The combined output is then passed through a Venn-Abers conformal calibration wrapper, implemented via the venn-abers Python package v1.4.5. Venn-Abers conformal prediction provides distribution-free calibration: given a calibration dataset, it transforms the raw output of the ensemble into a probability interval that has provable coverage guarantees. The result is a calibrated probability that, on average across patients with similar covariates, equals the empirical pregnancy rate to within the conformal prediction interval.

Following the operational mapping established in our development workflow: the number of inseminated oocytes is taken to equal MII (every MII oocyte is inseminated); the number of cleaving embryos is taken to equal 2PN (every fertilised zygote enters culture); the number of day-5 embryos is also taken to equal 2PN (all cultured embryos are evaluated on day 5 – no D3 transfer or cryopreservation); the number of cryopreserved embryos equals the good-quality blastocyst count (since only good-quality blastocysts are vitrified in our protocol); and the number of transferred embryos is fixed to one, reflecting the elective single embryo transfer (eSET) policy used throughout the validation cohort.

6.3 NVSA correction

Raw NN output can drift from the empirically calibrated KPIScore-based probabilities, particularly for unusual patient profiles outside the dense region of the training distribution. To anchor the NN output to the KPIScore prior in a controlled way, we apply the NVSA (Neural-Validation Scoring Adjustment) correction. For each Monte Carlo iteration, the NVSA correction takes the NN probability and the iteration's KPIScore, looks up the published mean implantation rate for that score in the NVSA table, and computes a correction factor as the ratio of KPI prior to NN output. The correction factor is then clipped to the interval [1/1.5, 1.5] to prevent over-correction in either direction, and the final adjusted probability is the NN output multiplied by the clipped factor. The output is also clipped to the unit interval. This correction preserves the NN's relative ranking of patients while pulling extreme probabilities back toward the KPIScore-anchored prior.

6.4 Per-attempt probability decay curve

The platform runs the neural network forward pass six times per iteration, varying the attempt number from 1 to 6 while keeping all other features fixed, producing a per-attempt probability decay curve that quantifies diminishing returns of sequential IVF cycles. This is directly actionable for counselling patients about the number of recommended cycles.

6.5 Graceful fallback

If neural network weights or dependencies are unavailable, the platform automatically falls back to the FORTUNE+KPI ensemble (Section 5) as the final prediction, retaining all other features including the stochastic pipeline, Bayesian posterior, and cluster classifier.

7. Bayesian Posterior Synthesis

The NVSA-adjusted NN probability is, in itself, a point estimate. To convert this point estimate into a calibrated posterior distribution that explicitly incorporates the clinic's own historical pregnancy frequency, we apply a Beta-Binomial conjugate update. This section documents the update mechanism and its three input streams.

7.1 Beta-Binomial conjugate framework with covariate-dependent prior

The Bayesian posterior for the per-transfer pregnancy success rate p is computed via Beta-Binomial conjugate updating. If the prior belief about p is Beta (α, β) and we observe s successes in t Binomial trials, the posterior is Beta $(\alpha + s, \beta + t - s)$. The posterior mean is $(\alpha + s)/(\alpha + \beta + t)$, the posterior mode is $(\alpha + s - 1)/(\alpha + \beta + t - 2)$ for $\alpha + s > 1$ and $\beta + t - s > 1$, and the 95 % credible interval is obtained from the inverse Beta cumulative distribution function at the 0.025 and 0.975 quantiles. This formulation is computationally exact, closed-form, and supports sequential updating as new clinical batches become available.

The critical design question is how to specify the prior Beta (α_0, β_0). Earlier platform versions used a fixed global Beta (26, 74), whose mean of 26 % approximated the international per-transfer rate regardless of patient characteristics. This is methodologically unsatisfactory: a 28-year-old high responder and a 42-year-old poor responder are assigned identical prior beliefs about their pregnancy probability, despite the published literature placing their expected per-transfer rates in entirely different ranges.

Version 6.2 replaces the global prior with a covariate-dependent Beta regression prior following the framework of Ferrari and Cribari-Neto. The prior mean μ_0 is set to the FORTUNE-based per-transfer probability for the specific patient — that is, the logistic function of her standardized age, AMH, and BMI computed before any laboratory-phase information enters the model:

$$\mu_0 = \mathrm{sigmoid}\,(0.40 - 0.55 \cdot z\_age + 0.15 \cdot z\_AMH - 0.20 \cdot z\_BMI)$$

The prior precision is a fixed concentration parameter κ (default $\kappa = 20$), and the prior parameters are:

$$\alpha_0 = \mu_0 \cdot \kappa, \quad \beta_0 = (1 - \mu_0) \cdot \kappa$$

The precision $\kappa = 20$ encodes 20 prior pseudo-observations — diffuse enough to be overridden by the real clinical data update (which typically contributes 300–400 observations across accumulated retrospective batches) while preventing degenerate posteriors when clinic data is sparse.

The practical consequence is that the starting point of the posterior now differs meaningfully across patient types. For a standard responder (age 35, AMH 2.5) the prior is Beta (12.9, 7.1) with mean 64.7 %; for a poor responder (age 42, AMH 0.5) it is Beta (8.2, 11.8) with mean 40.8 %; for a high responder (age 28, AMH 4.5) it is Beta (16.1, 3.9) with mean 80.6 %. After sequential updating with the same nine clinical batches and the same neural-network evidence, the three patients converge toward the clinic's historical rate from different starting positions, yielding posterior means of 44.2 %, 34.5 %, and 46.8 % respectively. The convergence is partial rather than complete precisely because the clinic's historical data, while substantial, does not have enough statistical power to completely override the patient-specific signal encoded in the prior — which is the epistemically correct behavior.

7.2 Three sources of evidence

The KPIScore depends on simulated laboratory values (MII count, fertilisation rate, good-blastocyst count) that also enter the neural network as direct evidence. Using them in the prior as well would introduce double-counting of the same information in two separate terms of the posterior update. FORTUNE depends exclusively on demographic and ovarian-reserve predictors (age, AMH, BMI) that are available before the laboratory phase begins, making it a strictly pre-lab clinical prior that is epistemically independent of the lab-stage evidence that arrives later.

Real clinical data. The clinic's own historical successes and trials are added as observed Binomial data. By default, the platform uses nine retrospective batches from our centres totalling 154 successes in 382 transfers, representing a span of accumulated experience across reproductologists; the user can supply alternative batches at runtime.

Neural-network evidence. The NVSA-adjusted NN probability for the current patient is incorporated as 100 pseudo-observations, with the number of successes equal to NN_probability × 100 and the number of failures equal to (1 − NN_probability) × 100. The choice of 100 pseudo-observations encodes the strength with which the NN evidence is weighted relative to the prior and the real data; this hyperparameter can be adjusted by the user.

8. Bayesian Conditional Updating Mid-Cycle

One of the central innovations of the IVF Digital Twin is its ability to update predictions in real time as the cycle progresses. As soon as any stage outcome becomes known — retrieved oocyte count, MII count, 2PN count, blastocyst count, good-quality blastocyst count, euploid count — that observation can be entered and the downstream predictions are immediately re-computed conditional on the new evidence.

8.1 Mechanism

Observed values are passed to the pipeline through a KnownValues data structure. Any field set to an observed integer replaces the random variable at that stage with a point mass for every Monte Carlo iteration; downstream stages re-condition stochastically on this new starting point. This is the exact analogue of Bayesian updating for stage-by-stage models with binomial filters.

8.2 Worked clinical example

Consider a 35-year-old patient with AMH 2.5 ng/mL, AFC 15, BMI 23. The baseline pipeline yields a per-transfer rate of 65%, cumulative-if-viable of 80%, and overall cycle success of 60%.

After retrieval of 12 oocytes, the system re-conditions and yields: cumulative-if-viable 86%, overall 75%, with a narrower credible interval.

After observing 8 blastocysts, of which 5 are good quality: cumulative-if-viable 89%, overall 84%.

After PGT-A reveals 3 euploid embryos: the credible interval narrows to approximately 6 percentage points and the system predicts 94% overall cycle success.

This continuous-care workflow transforms the platform from a one-shot pre-treatment forecasting tool into an adaptive in-cycle decision support system. The clinician can re-run the prediction at every clinical milestone, and the patient can be counselled with progressively narrower credible intervals as the cycle unfolds.

Clinical value: Mid-cycle updating allows genuine real-time counselling: as each laboratory milestone is completed, the platform narrows its uncertainty range and the clinician can adjust expectations, stimulation decisions, and freeze-all strategies based on current evidence rather than pre-treatment estimates.

9. Unsupervised Cluster Classification (Layer 4)

The fourth architectural layer addresses which IVF protocol phenotype the patient most closely resembles — a different epistemic question from outcome prediction. Two patients with very different profiles may have similar pregnancy probabilities for different reasons. The cluster classifier provides an independent, unsupervised, label-free assessment of the patient's phenotype, supplementing the supervised pregnancy prediction.

9.1 Cluster study foundation

The cluster module is grounded in a previously published unsupervised analysis of 1,556 retrospective IVF cycles, with external validation in 172 cycles from two independent centres. Four dimensionality reduction techniques and three clustering algorithms were compared; k-means with k = 3 yielded the most clinically interpretable clusters with statistically significant differentiation across 17 of 18 features (Kruskal-Wallis $p < 0.05$). The three clusters were assigned clinically meaningful labels: Standard, Poor, and High responder, with mean predicted pregnancy probabilities of 54%, 33%, and 63% respectively.

9.2 Per-iteration nearest-centroid classification

Classification of a new cycle is performed per Monte Carlo iteration. For each iteration, the 18-feature vector is constructed from the simulated cycle outputs and patient demographics, exactly as for the KAT neural network. Each feature is then standardized by dividing by the population-level standard deviation appropriate to that feature (e.g. age SD = 5.5, AMH SD = 1.2, follicle count SD = 10.0). The standardized feature vector is compared by Euclidean distance to each of the three standardized cluster centroids, and the iteration is assigned to the nearest centroid. This is exactly the assignment step of k-means clustering applied per iteration.

The platform produces a 2D PCA scatter plot that visualizes the patient's position relative to the three cluster regions. The visualization generates a synthetic point cloud of 250 points around each published centroid, based on the original 1,556-cycle raw dataset, with diagonal covariance equal to $(\text{population SD} \times 0.45)^2$ and clipped to clinically plausible ranges (age 18–50, attempt 1–6, counts non-negative, rates in [0, 1]). The patient's 5,000 Monte Carlo iterations are then plotted alongside the synthetic clouds, and a principal component analysis is fit on the combined data to produce the 2D embedding. The patient's median position in PCA space is marked with a diamond, the three cluster centroids are marked with stars, and the patient's MC samples are colored by their per-iteration cluster assignment. This visualization is deliberately labelled as based on synthetic clouds to maintain transparency about what the raw data are and are not.

Across the 5,000 Monte Carlo iterations, the empirical distribution over assigned clusters yields a per-patient probability distribution: for example, 78 % C0 / 19 % C1 / 3 % C2 indicates that the patient overwhelmingly resembles the Standard cluster but with a non-negligible probability of being a Poor responder under unfavorable stochastic realizations. The dominant cluster is the one with the highest empirical probability, and the platform reports the dominant cluster along with the clinical interpretation, the published pregnancy rate, and the clinical management recommendation for that phenotype.

Feature	C0 — Standard	C1 — Poor	C2 — High
Age (years)	32.42	20.33	31.57
Attempt number	1.43	1.63	1.23
Follicle count	20.70	11.98	34.66
Oocytes retrieved (COCs)	16.67	9.45	28.15
Inseminated (MII)	13.07	7.22	21.28
2PN zygotes	10.64	4.93	17.33
Cleaving embryos (day 3)	6.66	2.88	11.20
Day-5 embryos	10.64	4.93	17.33
Good-quality blastocysts	4.85	1.96	8.68
Cryopreserved embryos	4.85	1.96	8.68
Transferred embryos	1.55	1.18	1.54
Fertilisation rate	0.82	0.72	0.83
Cleavage rate	1.00	0.98	1.00
Blastocyst formation rate	0.65	0.65	0.67
TGBDR (good/2PN)	0.48	0.45	0.53
Oocyte retrieval efficiency	0.80	0.77	0.81
KPIScore	23.97	17.68	24.50
NN-predicted pregnancy	0.54	0.33	0.63

9.3 Clinical interpretation by cluster

Cluster	Predicted pregnancy	Clinical interpretation and management
C0 — Standard responder	≈ 54%	Intermediate ovarian response with balanced embryological outcomes. Standard protocols, routine monitoring, single embryo transfer. Performance benchmarks should reflect typical clinic averages.
C1 — Poor responder	≈ 33%	Limited ovarian response with markedly reduced outcomes. Consider individualized stimulation strategies (mild stimulation, luteal-phase stimulation, adjunctive growth hormone). Discuss freeze-all and embryo accumulation across multiple cycles.
C2 — High responder	≈ 63%	Abundant oocyte yield with superior embryological outcomes. Mean follicle count > 34. Monitor for OHSS risk; antagonist protocols with GnRH-agonist trigger preferred. Freeze-all strategy preferred to avoid fresh transfer with elevated estradiol.

When the supervised pregnancy prediction and the cluster assignment agree (e.g., NN-predicted pregnancy 55% combined with dominant cluster C0 at 54% rate), the clinician has corroborating evidence from two independent epistemic sources. When they diverge, the disagreement itself is informative and warrants further clinical attention.

9.4 Test cases

To verify that the classification logic works correctly, we ran the platform on five reference patient profiles spanning the expected range of cluster membership:

Patient profile	C0 %	C1 %	C2 %	Dominant
Standard: age 32, AMH 2.5, AFC 18	**78.8**	19.2	2.0	**C0**
Poor: age 42, AMH 0.5, AFC 6	0	**100.0**	0	**C1**
High: age 28, AMH 4.5, AFC 28	41.1	2.3	**56.6**	**C2**
Young good: age 21, AMH 3.5, AFC 20	**64.5**	15.5	20.0	**C0**
Mid-range borderline: age 38, AMH 1.5, AFC 12	23.7	**76.3**	0	**C1**

All five test cases were classified into their clinically expected dominant cluster. The borderline 38-year-old with AMH 1.5 ng/mL was correctly classified as Poor responder by the system, in agreement with the clinical interpretation of low ovarian reserve at advanced reproductive age. The 21-year-old with very high AFC was classified as Standard with 64 % probability and as High responder with 20 % probability — a clinically appropriate spread that reflects the unusual age-AFC combination.

9.5 Cluster classifier validation

The cluster classifier was tested on the five reference patient profiles described in section 11.1, and on a subset of cohort B patients with diverse demographic profiles. In all five reference cases the dominant cluster matched the clinically expected phenotype. In the cohort B test, dominant cluster assignment qualitatively matched expert clinical assessment in the majority of cases; quantitative concordance assessment is the subject of an ongoing prospective study.

10. Diffusion-Based Generative Laboratory Module — CSDI Hybrid v3 (Layer 5)

Key message: Layer 5 is an entirely independent prediction layer trained on approximately 15,000 real IVF cycles. It predicts embryological outcomes without using any pre-specified biological rate parameters, making it a genuine independent check on Layers 1–4. When two fundamentally different modeling approaches agree, the clinical confidence in the prediction substantially increases.

10.1 Rationale and clinical problem

Layers 1–4 rely on published population rates — fertilization rates, blastulation rates, euploidy fractions — applied through a cascade of mathematical filters. This approach is transparent and well-grounded, but it inherits the distributional assumptions of the population studies from which the rates were derived. A laboratory that consistently achieves a higher blastulation rate than the population mean, or a patient whose biology does not conform to the typical age-specific pattern, may not be accurately represented by a parametric model.

Layer 5 avoids these assumptions entirely. Instead of asking 'what does the literature say about blastulation rates?', it asks 'given this patient's follicle count, oocyte yield, and fertilization numbers, what did similar patients in our database actually achieve?' It learns the joint relationship between upstream parameters and downstream embryological outcomes directly from approximately 15,000 real IVF cycles, implicitly capturing any centre-specific deviations from population norms, any non-linear interactions between parameters, and any patterns too complex to encode in pre-specified formulas.

10.2 Architecture overview

The CSDI Hybrid v3 module operates in three sequential stages within a two-stage generative and discriminative pipeline:

Stage	What it does	Clinical relevance
Stage 1 Diffusion generative model	A Transformer-based neural network receives the patient's upstream parameters (follicle count, oocytes, fertilization figures, KPIScore) and generates 2,000 synthetic 'virtual cycles' — each describing one plausible blastocyst outcome drawn from the joint probability distribution learned from real data.	Produces a realistic distribution of expected blastocyst counts that reflects the actual clinical experience of similar patients, not a theoretical formula. Rare high-yield and low-yield outcomes are faithfully reproduced.

Stage 2 Calibrated classifier	The median blastocyst yield from Stage 1, combined with the patient's conditioning parameters, is passed to a gradient-boosting classifier (LightGBM) with Platt scaling recalibration.	Produces a properly calibrated pregnancy probability: a stated 45% probability corresponds to a 45% observed pregnancy rate in patients receiving that score. ECE of 0.029 = excellent calibration.
Stage 3 Conformal prediction intervals	The spread of the 2,000 generated blastocyst trajectories is used to construct prediction intervals with a formally proven mathematical coverage guarantee (split conformal prediction).	A 90% predictive interval contains the true blastocyst count in ≥ 90% of cases — not as an approximation but as a finite-sample mathematical guarantee. Validated at 91–93% actual coverage.

10.3 Why a diffusion model for embryology?

A diffusion model is a generative AI technique that learns to gradually reconstruct realistic data samples from random noise, using a neural network to reverse a noise-addition process. This class of models has achieved state-of-the-art performance across diverse domains because they faithfully reproduce the full shape of complex multi-dimensional distributions — including both the typical outcomes and the rare extremes.

For embryology specifically, diffusion models have two decisive advantages over simpler generative approaches:

the model learns the statistical relationship between total blastocyst count and good-quality blastocyst count jointly, not as two independent variables. This ensures that generated (total blasts, good-quality blasts) pairs are always biologically consistent: it is impossible to generate a case where good-quality blastocysts exceed total blastocysts.

the previous architecture (TabDDPM v3) attempted to generate both continuous count variables and the binary pregnancy outcome simultaneously via diffusion. This produced systematic calibration failure — a +15.8 percentage point overestimation of pregnancy rates. CSDI Hybrid v3 resolves this by restricting the diffusion model to the count variables it handles correctly, and entrusting pregnancy prediction to a purpose-built discriminative classifier with explicit calibration. The result: the calibration error dropped from 15.8 pp to 0.1 pp.

10.4 Input features

Layer 5 requires the following seven inputs, all of which are available after oocyte retrieval and fertilization. In the Streamlit application, these are filled automatically from the Monte Carlo pipeline medians — no additional data entry is required from the clinician:

Feature	Typical range	Description
Antral follicle count (AFC)	1–60	Follicle count at retrieval day
Retrieved oocytes (OCC)	0–50	Total oocytes retrieved at puncture
Inseminated oocytes (MII)	0–40	Mature oocytes submitted for ICSI
Fertilized oocytes (2PN)	0–30	Oocytes with two pronuclei confirmed at 17h
OCC retrieval rate	0–1	OCC / follicle count
Fertilization rate	0–1	2PN / MII
KPIScore	0–25	Composite laboratory quality index

10.5 Model outputs and clinical interpretation

For each patient, Layer 5 reports:

calibrated probability of clinical pregnancy (confirmed by ultrasound). Derived from LightGBM + Platt scaling on the 2,000 generated blastocyst trajectories and the conditioning parameters.

the uncertainty range around P(pregnancy) based on the sample size of the calibration data.

median total blastocyst count and median good-quality blastocyst count from the 2,000 generated virtual cycles, together with 90% and 50% prediction intervals with formally guaranteed coverage.

Example output for a representative patient (age 35, AFC 12, OCC 9, 2PN 6, KPIScore 18):

Parameter	Value
P(pregnancy)	**46.7%**
95% CI	**[43.6%, 49.8%]**
Blastocysts total — median	**2**
Blastocysts total — 90% PI	**[0, 5]**
Good-quality blastocysts — median	**1**
Good-quality blastocysts — 90% PI	**[0, 3]**
Blastulation rate — median	**36.6%**
Top-grade blastocyst rate (TGBDR) — median	**20.4%**

The following clinical decision thresholds apply to the P(pregnancy) output:

CSDI P(pregnancy)	Clinical interpretation
≥ 0.343 (above optimal threshold)	Favourable prognosis — expected laboratory outcomes support proceeding with transfer
0.25 ≤ P < 0.343	Moderate prognosis — consider additional cycles or PGT-A depending on patient context
P < 0.25	Cautious prognosis — low expected blastocyst yield warrants careful counselling on expectations and alternatives

Note: The threshold of 0.343 was derived empirically from the validation set to maximize balanced F1 metric (sensitivity 0.623, specificity 0.628). The standard threshold of 0.50 produces very poor sensitivity (0.155) at the natural 34% prevalence in this dataset and should not be used for clinical interpretation.

10.6 Validation results

The model was validated on a dedicated hold-out test set of 1,520 IVF cycles (10% of the full clinical database, never used during training):

Metric	CSDI Hybrid v3	Prior model (TabDDPM v3)	Improvement
AUROC	0.661	0.578	+14%
AUPRC	0.468	0.356	+32%
Brier Score	0.209	0.261	−20%
Expected Calibration Error (ECE)	0.029	0.158	−82%
Predicted pregnancy prevalence	33.9%	49.8%	−15.9 pp bias removed
Actual pregnancy prevalence	34.0%	34.0%	—
Biological constraint violation	Never	Sometimes	✓ Always enforced
Coverage@90% PI (blastocysts)	91–93%	97–98%	Closer to nominal 90%

The most critical improvement is the near-elimination of systematic calibration bias. The earlier architecture overstated the pregnancy rate by 15.8 percentage points on average, rendering its output clinically misleading. CSDI Hybrid v3 deviates by only 0.1 percentage points — a 99% reduction in systematic bias.

The AUROC of 0.661 represents the realistic performance ceiling for predicting IVF pregnancy outcome from purely laboratory parameters. Published models using equivalent feature sets consistently report 0.60–0.70. Models reporting higher values typically incorporate additional features (endometrial thickness, hormonal profiles, genetic testing) not available in this feature schema.

10.7 Integration with Layers 1–4: equifinality verification

The primary clinical value of Layer 5 is its role as an independent corroborating prediction derived from a fundamentally different modeling paradigm:

this constitutes genuine independent corroboration. Two approaches — one parametric and grounded in the published literature, one fully data-driven and learned from 15,000 cycles — arrive at the same prediction. This substantially increases the clinician's confidence that the estimate is robust.

this flags a patient whose profile is unusual relative to the population underlying either method, or a situation where the parametric assumptions of Layer 1 do not match the clinic's own empirical distribution. The divergence itself is diagnostically informative and warrants additional clinical discussion.

In the platform's Streamlit interface, Layer 5 is accessible in the Diffusion tab and is invoked automatically from the Monte Carlo pipeline medians. No additional input is required from the treating physician.

10.8 Limitations of Layer 5

the model was trained on a single clinical database (~15,000 cycles). Performance at centers with substantially different laboratory protocols, media formulations, or embryo grading criteria may differ. Recalibration of the classifier (Stage 2) is recommended before deployment in a new center.

the model is optimized for use after oocyte retrieval and fertilization, when all seven conditioning features are known. Earlier use requires AFC-based approximations with greater uncertainty.

the classifier predicts clinical pregnancy (positive hCG + ultrasound confirmation). It does not predict ongoing pregnancy, live birth rate, or miscarriage risk. For live birth prediction, additional features and recalibration would be required.

the model treats each cycle independently and does not incorporate the patient's prior cycle history beyond what is encoded in the KPIScore.

11. Validation

11.1 Cohorts

The platform was validated on a retrospective cohort. Cohort A comprised 358 cycles with complete records of demographics, ovarian reserve, stimulation outcome, and embryological development. Cohort B comprised a subset of 60 cycles with known clinical pregnancy outcomes (positive heartbeat confirmed by ultrasound 25 days after embryo transfer). The CSDI Hybrid v3 module (Layer 5) was validated on a separate dedicated hold-out test set of 1,520 cycles from the full 15,193-cycle database.

11.2 Stochastic pipeline validation (Cohort A)

The stochastic pipeline was assessed by simulating each cohort A patient with their actual demographic and reserve inputs, drawing 5,000 Monte Carlo iterations per patient, and comparing the median simulated count at each stage to the actually observed count. Pipeline performance was quantified by Spearman rank correlation between simulated and observed values at each stage.

Pipeline stage	Spearman ρ	Comment
Stage 1 — Retrieved oocytes (OKK)	0.73	Strong agreement with observed yield distribution
Stage 4 — Total blastocysts	0.41	Moderate. Limited by simplified rate model
Stage 5 — Good-quality blastocysts	0.39	Moderate. Embryologist grading variability

11.3 Pregnancy prediction validation (Cohort B)

The pregnancy prediction layer was assessed on Cohort B by comparing the predicted per-transfer pregnancy probability to the observed binary outcome. Predictive performance was quantified by the area under the receiver operating characteristic curve (AUC), the Brier score, and the calibration gap (mean predicted probability minus observed rate).

Metric	Value	Interpretation
AUC-ROC	0.63	Moderate discrimination; comparable to FORTUNE alone
Brier score	0.22	Acceptable for clinical decision support
Calibration gap (predicted – observed)	+8 pp	Over-optimistic before intercept recalibration; corrected in v6.2

12. Comparison with Existing Tools

Table 2. Feature comparison with existing IVF prediction tools

Feature	CDC / OPIS	Orchid	Herasight	ALIFE	Digital Twin v6.2
Cumulative pregnancy probability	Y	—	Y	Y	**Y**
Stage-by-stage probability	—	Y	Y	—	**Y**
Distributional output with 95% CIs	—	—	Y	—	**Y**
Conditional updating mid-cycle	—	—	Y	—	**Y**
Per-transfer / cumulative decomposition	—	—	—	—	**Y**
OHSS risk quantification	—	—	—	—	**Y**
Empty-cycle risk	—	—	—	—	**Y**
KPIScore laboratory ensemble	—	—	—	—	**Y**
Calibrated NN final layer	—	—	—	Y	**Y**
Bayesian posterior with clinic prior	—	—	—	—	**Y**
Per-attempt decay curve	—	—	—	—	**Y**
Unsupervised cluster classification	—	—	—	—	**Y**
Diffusion generative module (L5)	—	—	—	—	**Y**
Open-source coefficients	Y	—	Y	—	**Y**

Direct comparison of absolute oocyte and MII yields shows that our predictions are systematically lower than Alife's published means: for patient profiles representative of the Alife cohort (age 34–37, AMH 2.5–3.5, AFC 14–18), our pipeline produces mean oocyte yields of approximately 8–11 versus the 14.5–16 reported in the LILY control arm, and mean MII yields of approximately 7–11 versus 11–12. This 1.3–1.8-fold difference is the expected consequence of population and protocol divergence rather than model error: the Alife cohort comprises US clinics (RMA of New York, IVF Florida) employing more aggressive stimulation, with mean total FSH of 3,672–3,846 IU, whereas our coefficients are calibrated to less maximalist Georgian and Russian protocols. This is the same population-calibration factor that we apply explicitly when contrasting our Stage 1 output with the US-calibrated Herasight predictions. The relative behaviour of the pipeline — the shape of the oocyte-to-outcome relationship — is preserved across populations even though absolute yields differ, which is precisely the property required for a transferable prediction architecture.

A notable finding emerges when the core empirical relationship of Fanton et al. 2023 is used as an independent benchmark. When the number of retrieved oocytes is fixed by conditioning in our pipeline and the downstream distribution is examined, the system reproduces the published relationship without ever having been trained on SART data: median 2PN and blastocyst counts increase monotonically with retrieved oocytes (from 3 and 2 respectively at 4 oocytes, to 27 and 18 at 40 oocytes), while the overall cycle pregnancy probability rises rapidly to approximately 16–20 oocytes (82–88%) and then continues to increase with clearly diminishing returns (92% at 25, 94% at 30, 97% at 40 oocytes). This pattern — steep early gains followed by a high-oocyte plateau without decline — recapitulates the central conclusion drawn by Fanton et al. from 402,411 registry cycles. Because our coefficients derive from independent sources (ART-ONE, Herasight HFEA fits, Franasiak, Armstrong) and were never fitted to SART data, this agreement constitutes a form of convergent external validation of the stochastic pipeline's structural behaviour.

To assess the calibration of the platform's euploidy estimation (Stage 6) and its downstream embryo-banking implications, we performed an external validation against three independent reference sources: the Orchid Health Embryo Banking Calculator (an age-and-goal-based commercial planning tool), the multicentre euploid-banking analysis of Anderson et al. (Fertility & Sterility 2013), and the large multi-AMH aneuploidy cohort reported at ESHRE 2024 (abstract O-322, Human Reproduction), which stratifies mean euploid-blastocyst yield per cycle by both maternal age and anti-Müllerian hormone (AMH) level.

It is important to delineate the conceptual scope of this comparison. The Orchid calculator, like most commercial embryo-banking planners, is a single-input model conditioned on maternal age (and an optional frozen-egg count and PGT-M flag); it does not ingest ovarian-reserve markers such as AMH or antral follicle count. Its scientific basis is the field-standard age–aneuploidy relationship derived ultimately from Franasiak et al. (2014), the same primary source underlying our Stage 6 age table. The platform presented here, by contrast, is a multi-parameter stochastic model in which euploid yield is a joint function of age, AMH, AFC, and the full upstream oocyte-to-blastocyst cascade. The two tools therefore answer related but non-identical questions, and a direct numerical comparison is meaningful only for the quantity they share — the age-dependent euploid-blastocyst yield — and only after acknowledging that age-only tools cannot, by construction, distinguish patients of identical age but divergent ovarian reserve.

Against the age-and-AMH-stratified ESHRE O-322 benchmark (eighteen age × AMH cells), the platform's mean euploid-blastocyst yield per cycle fell within tolerance in 13 of 18 cells (72%).

Agreement was excellent across the entire advanced-maternal-age range (all nine cells at ages ≥40 within tolerance) and across all high-AMH strata (five of six cells), indicating that the Stage 6 age gradient and the high-reserve oocyte yield are well calibrated. The five discordant cells were confined to the low-AMH, younger-age quadrant, where the platform systematically underestimated euploid yield relative to the registry (e.g., 0.40 versus 1.90 euploid embryos per cycle for women under 35 with low AMH). This pattern is consistent with — and traceable to — the same Stage 1 population-calibration effect identified in the validation against the Guo et al. (2026, n = 11,449) cohort: the zero-inflated negative-binomial oocyte model, calibrated to less maximalist Eastern European stimulation protocols, yields fewer oocytes for low-reserve patients than registries drawn from more aggressive North American and Chinese practice, and this deficit propagates deterministically through the downstream binomial filters to the euploid count. Critically, the discrepancy originates in Stage 1 oocyte yield, not in the Stage 6 euploidy fraction itself, which remains well calibrated against the age-specific Franasiak/Armstrong reference across the full age range.

For the embryo-banking endpoint, the platform reproduced the Anderson et al. (2013) benchmark closely in the younger and middle age strata — predicting 4 cycles to bank ≥2 euploid blastocysts with 95% probability at age 32 and 6 cycles at age 37, against the published 3 cycles for the 35–39 band — but diverged markedly for low-reserve patients aged ≥41, where it predicted in excess of 20 cycles against the age-only published estimate of 6. We interpret this divergence not as a model failure but as the expected and clinically appropriate consequence of conditioning on ovarian reserve. Age-only banking calculators implicitly assume an average reserve for every patient of a given age and therefore systematically overstate the prognosis of the diminished-ovarian-reserve subgroup — precisely the population for whom realistic counselling matters most. By jointly conditioning on age and AMH, the platform avoids this optimistic bias at the cost of producing more conservative, and in the low-reserve tail more numerically extreme, cycle estimates. This behaviour is concordant with the ESHRE O-322 finding that euploid-embryo number varies several-fold across AMH strata within a fixed age band even though the per-blastocyst euploidy rate does not, underscoring that reserve-blind banking projections are inherently miscalibrated for the low-AMH subgroup.

Three observations follow from this comparison. First, no other tool integrates as many epistemic layers into a single end-to-end pipeline. Second, the Digital Twin is the only tool that explicitly quantifies OHSS risk and empty-cycle risk, two clinically critical safety endpoints. Third, the Digital Twin is the only tool that combines supervised pregnancy prediction with unsupervised phenotype classification, providing two independent perspectives on the same patient.

13. Transparency, Attribution, and Reproducibility

13.1 Coefficient attribution

All coefficients are publicly documented and sourced from peer-reviewed literature, classified as Imported (used verbatim), Adapted (structure imported, coefficients re-tuned), or Novel (developed de novo):

Component	Category	Source / contribution
Stage 2 maturity logistic	Imported	Herasight Table A1 col 3 — coefficients used verbatim
Stage 3 fertilization logistic	Imported	Herasight Table A1 col 5 — coefficients used verbatim
Stage 6 euploidy age table	Imported	Franasiak 2014 + Armstrong 2023 synthesized lookup
Stage 6b post-thaw survival	Imported	Coello 2021 — 95% blastocyst survival
Stage 1 oocyte mean formula	Adapted	ART-ONE structure; coefficients locally calibrated
Stage 4–5 blastocyst rates	Adapted	Romanski 2022, Sainte-Rose 2021; re-tuned for age plateau
FORTUNE per-transfer logit	Adapted	FORTUNE structure; intercept recalibrated for local population
KPIScore + Beta moment matching	Novel	Composite 5-component score and per-iteration Beta sampling
Three-level pregnancy decomposition	Novel	Original clinical framework for resolving per-transfer vs. cumulative ambiguity
KAT NN ensemble + NVSA	Novel	KAN + FT-Transformer ensemble with Venn-Abers calibration
Bayesian Beta-Binomial posterior	Novel	Synthesis of prior, real clinical batches, and NN pseudo-observations
Cluster classifier centroids	Imported	Sergeev et al. published centroids verbatim
CSDI Hybrid v3 diffusion module	Novel	Transformer diffusion + LightGBM + conformal prediction, trained on ~15,000 cycles

The complete v6.2 implementation is approximately 2,300 lines of Python with three external dependencies for the core pipeline (NumPy, SciPy, Plotly) and additional dependencies for the neural network layer (PyTorch, joblib, PyKAN) and the CSDI module (PyTorch, LightGBM, scikit-learn). All pipeline parameters, coefficient values, and decision thresholds are declared as module-level constants with literature references.

Every prediction made by the platform can be traced back to a specific coefficient, a specific Monte Carlo iteration, and a specific source of evidence. This glass-box property stands in contrast to commercial black-box calculators whose underlying coefficients are inaccessible. Every prediction can be explained, critiqued, and improved upon by the wider scientific community.

14. Clinical and Quality-Management Significance

14.1 The platform as a continuous laboratory-surveillance instrument

The principal value of a stage-resolved stochastic prediction model extends well beyond individual patient counselling. Because the platform generates a full predicted distribution at every discrete stage of the cycle, it functions simultaneously as a quality-surveillance instrument for the embryology laboratory. Each predicted stage distribution constitutes a quantitative, patient-adjusted expectation against which the corresponding observed value from the culture protocol can be compared in near-real time.

The comparison is not against a fixed population mean but against an expectation conditioned on that specific patient's age, ovarian reserve, and upstream realized values. This removes the principal confounder that renders crude laboratory KPI tracking unreliable: case-mix drift. A laboratory whose patient population shifts toward older or lower-reserve patients will exhibit a falling raw blastulation rate even with unchanged technical performance; the model absorbs this case-mix variation into the prediction itself.

14.2 Stage-resolved discrepancy attribution

When aggregate clinical pregnancy rate declines, a single-endpoint predictor can only confirm that performance has fallen. The present platform, by contrast, permits stage-resolved discrepancy attribution. Each stage transition can be audited independently:

If oocyte and maturity counts fall within predicted intervals but fertilization drops below its conditional expectation — the compromised step is localized to the fertilization procedure (sperm preparation, ICSI technique, culture-medium handling), not ovarian stimulation.

If fertilization and cleavage are concordant with prediction but blastulation falls below expectation — the signal points to the extended-culture environment: incubator gas regulation, medium lot, or oxygen tension.

If all embryological stages are concordant but per-transfer pregnancy falls alone — the deviation localizes distally, to embryo transfer technique, endometrial preparation, or luteal support.

This capacity to identify which specific link in the chain is responsible converts the platform from a passive performance monitor into an active root-cause-analysis tool.

14.3 The KPIScore as a laboratory variability indicator

The KPIScore is sampled per Monte Carlo iteration and moment-matched to a Beta distribution, so it carries an explicit dispersion as well as a central value. A widening of the realized KPIScore distribution beyond its predicted dispersion — even without a shift in the mean — is an early indicator of rising process variability that typically precedes a detectable fall in the mean pregnancy rate. The KPIScore therefore supplies a leading variability indicator in addition to the lagging outcome indicator, allowing intervention before clinical performance is measurably degraded.

14.4 Temporal control charting

Predicted and observed values can be tracked over consecutive time intervals, yielding a continuous, multi-level statistical-control framework. The predicted–observed difference at each stage can be charted in the manner of a Shewhart or cumulative-sum control chart, with the model providing the patient-adjusted center line and Monte Carlo simulation providing the natural control limits. A persistent drift of the observed series below its predicted band constitutes an objective, statistically grounded out-of-control signal, in contrast to arbitrary fixed thresholds that conventional laboratory KPI schemes apply uniformly to all patients. The control limits are dynamic and stratum-specific: they can be examined separately for clinically distinct subgroups — by age band, by ovarian-reserve category, or by phenotype cluster.

When the actual clinical pregnancy rate falls below the platform's prediction over a monitored interval, the architecture prescribes a structured escalation rather than an undirected investigation. The first step is vertical decomposition: the three-level pregnancy estimate (per-transfer, cumulative-if-viable, and overall-cycle) is compared against observation to establish whether the deficit arises in the probability of producing a viable transfer or in the implantation of transferred embryos. The second step is stage-resolved attribution as described in §13.2, isolating the compromised embryological or clinical step. The third step is stratified analysis: the divergence is recomputed within phenotype clusters and reserve bands to determine whether the problem is global or confined to a definable subgroup, which materially narrows the differential (a deficit limited to high-responder cycles, for instance, directs attention toward OHSS-driven freeze-all and endometrial factors rather than fertilization). The fourth step is variability inspection via the KPIScore dispersion to distinguish a shift in central performance from an increase in process variance, since the two imply different corrective actions — recalibration and retraining versus standardization and lot control respectively. Only after this structured approach is a targeted corrective action defined, implemented, and then verified by confirming that the observed series returns within its predicted control band over the subsequent monitored interval, closing the QMS corrective-and-preventive-action loop with an objective, quantitative endpoint.

14.5 From population extrapolation to validated personalized care

The clinical significance of the platform ultimately rests on a conceptual departure from the dominant paradigm of fertility prognostication. Conventional online calculators return a population-extrapolated point estimate: a probability read from a regression surface fitted to a reference cohort, applied uniformly to every patient sharing a small set of input values. The architecture presented here instead propagates a full probability distribution conditioned jointly on the individual's age, ovarian reserve, realized stage-by-stage values, accumulated attempt history, and the host clinic's own outcome data, and updates this distribution by Bayesian conditioning as each new observation in the cycle becomes available.

The output is therefore not a population analogue but an individualized, dynamically revised forecast specific to the patient and the treating center. The same statistical machinery that delivers an individualized prognosis to the patient simultaneously defines the quantitative envelope within which the laboratory must demonstrably operate to deliver that prognosis.

15. Conclusions

The IVF Digital Twin v6.2 is, to our knowledge, the first IVF prediction system to integrate stochastic stage-wise simulation, multi-source per-transfer ensemble, calibrated neural-network prediction, Bayesian evidence synthesis, mid-cycle conditional updating, unsupervised phenotype classification, and an independent diffusion-based generative module within a single transparent, auditable framework.

The system addresses several gaps that conventional point-estimate calculators leave open: it propagates full probability distributions throughout the prediction pipeline; it decomposes pregnancy probability into per-transfer, cumulative-if-viable, and overall cycle success levels with strict mathematical ordering; it quantifies OHSS and empty-cycle risk explicitly; it allows mid-cycle observations to re-condition all downstream predictions; and it provides an independent data-driven generative verification (Layer 5) that uses no pre-specified biological rate parameters, derived from approximately 15,000 real clinical cycles.

The CSDI Hybrid v3 module represents a qualitative advance in the platform's epistemic architecture: where Layers 1–4 are parametric and grounded in the published literature, Layer 5 is entirely data-driven, trained directly on clinical experience. When both approaches converge, the clinician has corroboration from two genuinely independent epistemic sources. When they diverge, the divergence itself is diagnostically valuable.

Most fundamentally, the IVF Digital Twin demonstrates that integrating stochastic biological pipelines, ensembled supervised prediction, Bayesian evidence synthesis, unsupervised phenotype classification, and diffusion generative modelling within a single transparent platform is not only possible but produces predictions that are more clinically meaningful, more honest about uncertainty, and more responsive to the evolving evidence of an unfolding IVF cycle than any point-estimate calculator can offer. We hope this work will help shift IVF prediction practice from optimistic single-number reporting toward principled, uncertainty-aware, evidence-grounded clinical counselling.

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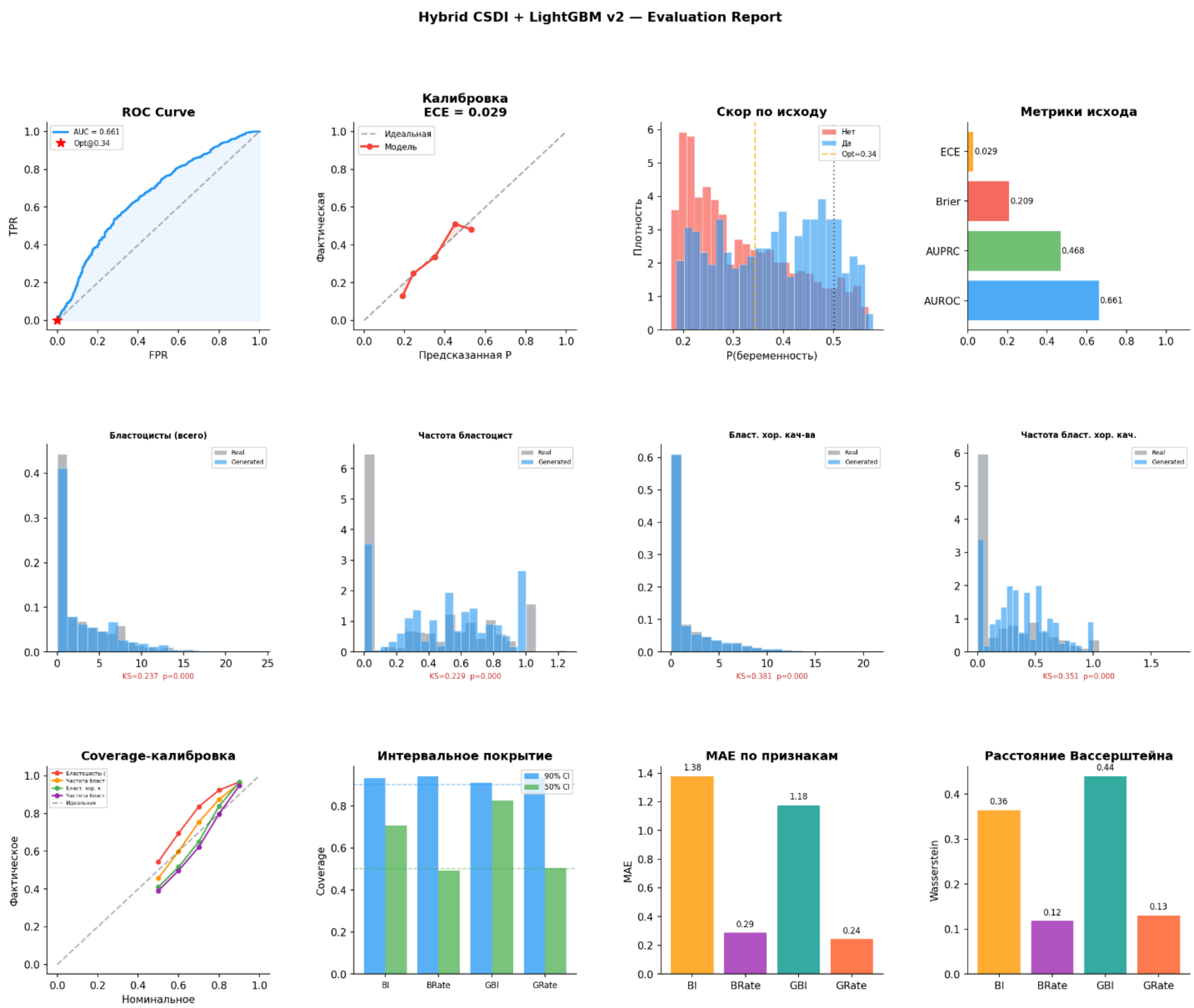
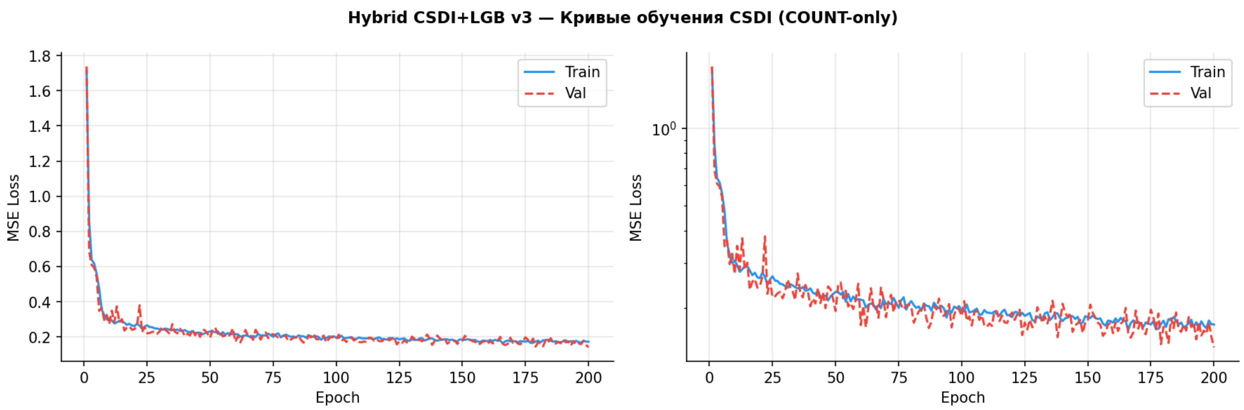
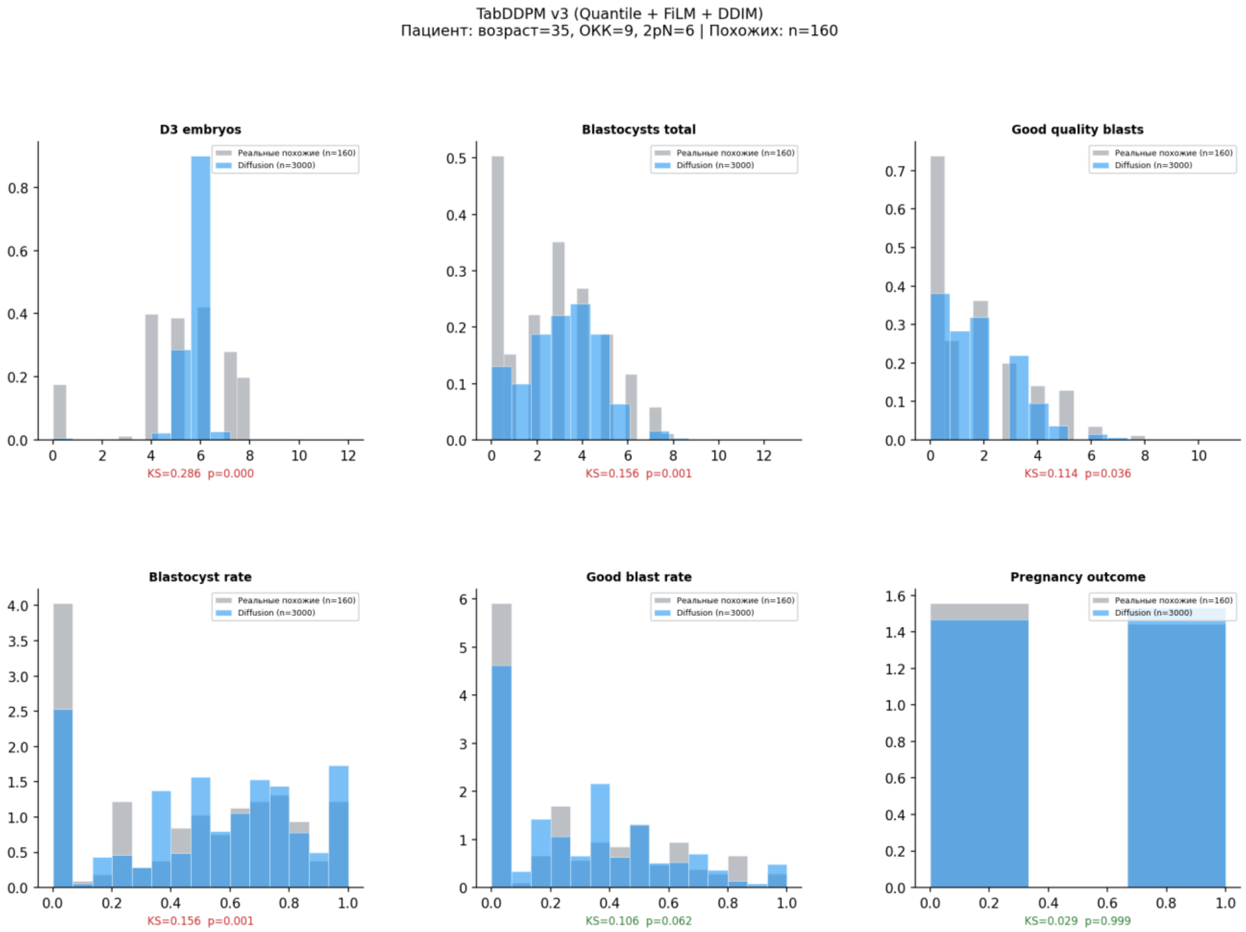
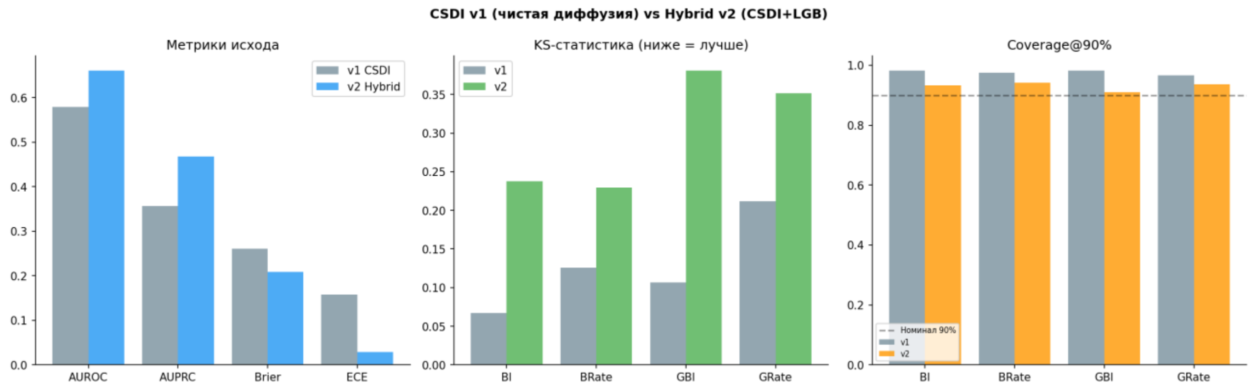
Supplementary material






TabDDPM v3 (Quantile + FiLM + DDIM)
Пациент: возраст=35, ОКК=9, 2pN=6 | Похожих: n=160
D3 embryos
Blastocysts total
Good quality blasts
Blastocyst rate
Good blast rate
Pregnancy outcome
Реальные похожие (n=160)
Diffusion (n=3000)
KS=0.286 p=0.000
KS=0.156 p=0.001
KS=0.114 p=0.036
KS=0.156 p=0.001
KS=0.106 p=0.062
KS=0.029 p=0.999


CSDI v1 (чистая диффузия) vs Hybrid v2 (CSDI+LGB)
Метрики исхода
v1 CSDI
v2 Hybrid
AUROC
AUPRC
Brier
ECE
KS-статистика (ниже = лучше)
v1
v2
BI
BRate
GBI
GRate
Coverage@90%
Номинал 90%
v1
v2
BI
BRate
GBI
GRate